

SDN MULTI-TENANCY TRAFFIC ISOLATION

Implementation Strategies Using OpenFlow in Shared Data Center Infrastructure

OVERVIEW

In a multi-tenant data center sharing physical infrastructure, an SDN controller uses OpenFlow to enforce complete traffic isolation across tenant workloads. Rather than relying solely on traditional static VLANs, an SDN controller programmatically programs flow tables across the physical switches to dictate exact isolation, encapsulation, and routing behaviors.

1. PRIMARY OPENFLOW MECHANISMS FOR MULTI-TENANCY

- **Flow-Table Match-Action Rule Enforcement:** The SDN controller programs unique flow entries matching tenant-specific identifiers (e.g., Ingress Port, Source/Destination MAC addresses, VLAN tags, or Tunnel IDs). Any traffic that does not explicitly match a tenant's programmed flow rules hits a default "Drop" rule, blocking inter-tenant communication at the hardware level.
- **Virtual Network Encapsulation (VXLAN / NVGRE Integration):** OpenFlow actions can push, pop, or rewrite encapsulation headers such as VXLAN Network Identifiers (VNIs) or VLAN IDs at ingress and egress edge switches. This encapsulates tenant frames inside L3 IP tunnels, providing up to 16 million distinct, isolated virtual networks across shared physical switches.
- **Slicing and Flow Hypervisors:** Utilizing OpenFlow virtualization layers (such as FlowVisor or OpenVirteX), the controller partitions physical switch resources into distinct logical topologies. Each tenant can be assigned an isolated control plane slicing context, ensuring tenant rules cannot read or modify other tenants' flow spaces.
- **Strict Group and Meter Enforcements (QoS Isolation):** OpenFlow Metering and Group Tables allow the controller to enforce bandwidth caps and rate-limiting per tenant. This prevents noisy-neighbor scenarios where one tenant's traffic burst starves another tenant's network performance.

2. IMPLEMENTATION WORKFLOW

- **Ingress Tagging:** As traffic enters an edge switch port assigned to Tenant A, the OpenFlow controller installs a flow rule that tags or encapsulates the packet with Tenant A's unique identifier (e.g., VNI 1001).
- **Core Forwarding:** Intermediate switches forward traffic strictly based on the match criteria containing the tenant identifier and destination address.
- **Egress Decapsulation:** At the egress switch port connected to Tenant A's destination host, the outer tenant encapsulation is stripped before the frame is delivered.
- **Inter-Tenant Blocking:** If Tenant A attempts to communicate with Tenant B, the flow table lookup fails to match any valid forwarding rule, triggering an explicit DROP action or forwarding to the controller for dropping.

CONCLUSION

By combining match-action flow entries, dynamic overlay encapsulation (VXLAN/NVGRE), and default-deny policies, an OpenFlow SDN controller enforces strict, scalable, multi-tenant isolation across shared physical switch infrastructure.